

# CURRICULUM VITAE

McMaster University  
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## EMPLOYMENT

- 2025 – 2026 **McMaster University** Britton Postdoctoral Fellow  
RESEARCH GROUP: Combinatorial Algebraic Geometry  
2025 Spring **Fields Institute for Research in Mathematical Sciences** Postdoctoral Fellow  
THEMATIC PROGRAM: Commutative Algebra and Applications  
2024 Fall **Max Planck Institute for Mathematics in the Sciences** Postdoctoral Fellow  
RESEARCH GROUP: Nonlinear Algebra

## EDUCATION

- 2024 Ph.D. **University of Minnesota, Twin Cities** ADVISOR: Christine Berkesch  
THESIS: Diagonalization, Direct Summands, and Resolutions of the Diagonal  
2017 B.A. **University of California, Berkeley** (with Honors) ADVISOR: David Eisenbud  
THESIS: Local Computations in Macaulay2  
2016 **Math in Moscow** (semester abroad at the Independent University of Moscow)

## AWARDS

- 2023 – 2024 Doctoral Dissertation Fellowship, University of Minnesota  
2020 Honorable Mention, Graduate Research Fellowships Program, NSF  
2018 – 2020 John Ordway Fellowship, School of Mathematics, University of Minnesota

## GRANTS

- 2025 NSF DMS-2508868 for Macaulay2 Workshop at UW-Madison  
2023 NSF DMS-2302476 for M2Week at the University of Minnesota  
2022 NSF DMS-2206872 for GradMoCCA at the University of Minnesota  
2018 DRP Network Seed Grant

## PUBLICATIONS

- 2025 9. “Computing Direct Sum Decompositions”  
D. Mallory. *Journal of Symbolic Computation* **133**, March (2026). [arXiv:2412.19799](#)  
2024 8. “A short resolution of diagonal for smooth projective toric varieties of Picard rank 2”  
M. K. Brown. *Algebra & Number Theory* **18-10** (2024), 1923-1943.  
2020 7. “The virtual resolutions package for Macaulay2”  
A. Almousa, J. Bruce, and M. C. Loper. *J. of Software for Algebra & Geometry* **10** (2020), 51–60.  
2017 6. “What is the optimal way to prepare a Bell state using measurement and feedback?”  
L. Martin and K. B. Whaley. *Quantum Sci. & Technol.* **2** (2017), no. 4.

## PREPRINTS

- 2025 5. “Computing global Ext for complexes” [arXiv:2509.25103](#)  
M. Brown, S. Dey, G. Li. Submitted.  
2025 4. “Connection Matrices in Macaulay2” [arXiv:2504.01362](#)  
P. Görlach, J. Koefler, A.-L. Sattelberger, H. Schroeder, N. Weiss, F. Zaffalon. Submitted.  
2024 3. “Splitting of vector bundles on toric varieties”  
Submitted. [arXiv:2412.19793](#)  
2022 2. “Bounds on multigraded regularity”  
J. Bruce, and L. Cranton Heller. Submitted. [arXiv:2208.11115](#)  
2021 1. “Characterizing multigraded regularity and virtual resolutions on products of projective spaces”  
J. Bruce, and L. Cranton Heller. Submitted. [arXiv:2110.10705](#)

## INVITED TALKS (selected)

### Conferences

- 2026 Jan. Joint Math Meetings, Washington, D.C.
- 2025 Sep. Route 81 Mathematics Conference, Queen's University, Kingston, Canada
- 2024 July Effective Methods in Algebraic Geometry, MPI MiS, Leipzig, Germany
- May Spring Western AMS Sectional Meeting, San Francisco State University
- Jan. Joint Math Meetings, San Francisco, California
- 2023 Oct. Fall Central AMS Sectional Meeting, Creighton University, Omaha
- Jul. SIAM Applied Algebraic Geometry Conference, Eindhoven University, the Netherlands
- Jul. Géométrie Algébrique en Liberté, University of Warwick, UK
- Apr. CA+ Conference, University of Minnesota, Twin Cities
- Mar. Spring Southeastern AMS Sectional Meeting, Georgia Tech, Atlanta
- 2022 Oct. Fall Western AMS Sectional Meeting, University of Utah, Salt Lake City
- Sep. Fall Central AMS Sectional Meeting, University of Texas, El Paso
- May Macaulay2 Conference, Cleveland State University
- 2021 Apr. Spring Central AMS Sectional Meeting, University of Cincinnati (virtual)
- 2019 Aug. SIAM Applied Algebraic Geometry Conference, Universität Bern, Switzerland
- 2018 Jan. Joint Mathematics Meetings, San Diego, California

### Seminars

- 2025 Oct. Syracuse University, Algebra Seminar
- Jun. Fields Institute, Commutative Algebra and Applications Seminar
- 2024 Dec. Technische Universität Berlin, Discrete Geometry Seminar
- Dec. Universität Osnabrück, Algebra Seminar
- Dec. Universität Bielefeld, BIREP Seminar
- Nov. University of Edinburgh, Hodge Institute, EDGE Geometry Seminar
- Oct. Friedrich-Schiller-Universität Jena, Algebra Seminar
- May University of Oregon, Eugene, Algebra Seminar
- Apr. UC Berkeley, Commutative Algebra and Algebraic Geometry Seminar
- Mar. University of Michigan, Ann Arbor, Commutative Algebra Seminar
- Feb. Georgia Tech, Atlanta, Algebra Semina
- Feb. Washington University in St. Louis, Combinatorics Seminar
- 2023 Nov. University of Notre Dame, Algebraic Geometry and Commutative Algebra Seminar
- Apr. University of Nebraska - Lincoln, Commutative Algebra Seminar
- Feb. Texas A&M University, Algebra and Combinatorics Seminar
- Jan. University of Illinois, Chicago, Commutative Algebra Seminar
- 2022 Oct. University of Utah, Algebraic Geometry Seminar
- Oct. Arizona State University, Number Theory and Algebra Seminar
- 2021 Oct. Georgia Tech, Algebra Seminar
- Oct. Auburn University, Algebra Seminar
- Feb. ICERM semester in Combinatorial Algebraic Geometry, Grad/Postdoc Seminar (virtual)

### Posters

- 2024 Oct. Group actions, combinatorics, and Fano varieties, Universität Innsbruck, Austria
- Sep. Syzygies and Hilbert Schemes, Jagiellonian University, Kraków, Poland
- Jan. Connections Workshop: Commutative Algebra, SLMATH, Berkeley
- 2023 Nov. Western Algebraic Geometry Symposium, Washington University in St. Louis
- May MSRI/SLMATH Summer School in Commutative Algebra, University of Notre Dame
- 2022 Nov. Western Algebraic Geometry Symposium, UC Riverside
- Jun. Pan-American School in Commutative Algebra, CIMAT, Mexico
- 2017 Aug. SIAM Applied Algebraic Geometry Conference, Georgia Tech, Atlanta

## TEACHING and MENTORING EXPERIENCE

### Department of Mathematics & Statistics, McMaster University

2026 Winter MATH 1MM3: instructor of record for Applied Calculus.

2025 Fall MATH 3GR3: instructor of record for Abstract Algebra (class of about 40 students).

### School of Mathematics, University of Minnesota, Twin Cities

2023 Spring MATH 1272: instructor of record for Calculus II (class of about 100 students).

2022 Fall MATH 8253: graded homework for Algebraic Geometry (2nd year graduate course).

2022 Spring MATH 1272: instructor of record for Calculus II (class of about 90 students).

2019 – 2021 MATH 2243: TA for Linear Algebra & Differential Equations (six sections total).

2018 Fall MATH 1271: TA for Calculus I (two sections total).

2020 Summer REU: mentored P. Cranford, A. Peng, and V. Srinivasan towards [arXiv:2106.12667](#).

2019 & 2022 DRP: mentored five undergraduate students in weekly reading projects.

### Department of Mathematics, UC Berkeley

2015 Spring Math113: graded homework for Abstract Algebra.

2014 Fall Math116: graded homework and assisted the instructor for Mathematical Cryptography.

### Math Center, Irvine Valley College

2013 – 2014 Tutored lower-div courses including linear algebra, differential equations, and discrete math.

### Mathobotix, Irvine, CA

2012 – 2014 Developed curriculum for computer science-based problem-solving using Python.

## CONFERENCE AND WORKSHOP ORGANIZATION

2025 Dec. **Special Session at the Canadian Mathematical Society Winter Meeting**

Special session in commutative algebra.

2025 Jul. **Macaulay2 Workshop in Madison**

Five-day workshop prior to SIAM-AG 2025 with 55 participants in Madison.

2024 Nov. **Macaulay2 in the Sciences Workshop**

Three-day workshop on computational algebra with 45 participants in Leipzig, Germany.

2023 Jun. **Macaulay2 Workshop & Mini-school**

Weeklong event in computational commutative algebra bringing 70 researchers to Minneapolis.

2022 May **Graduate Meeting on Combinatorial Commutative Algebra**

Weekend graduate event with 12 speakers bringing 70 students and postdocs to Minneapolis.

## MATHEMATICAL SERVICE

2018 – 2023 **Directed Reading Program**, University of Minnesota, Co-founder and co-organizer  
Matched over 250 undergraduate and graduate students in guided mathematics reading projects.

2019 – 2022 **Student Commutative Algebra Meeting**, University of Minnesota, Co-organizer  
Held a weekly meetup of friendly neighborhood commutative algebra students.

2021 Apr. **Graduate Student Combinatorics Conference**, Session Chair  
Chaired the session on Combinatorial Algebraic Geometry.

2017 – 2018 **Pauline Sperry Undergraduate Lecture Series**, UC Berkeley, Organizer  
Inaugurated an annual lecture aimed at providing a role model for marginalized students in math.

### Referee Work:

Journal of Pure and Applied Algebra, Journal of Software for Algebra and Geometry

## BROADER OUTREACH

2019 – 2023 **Vincent Hall Thespians, University of Minnesota**, Performer  
Helped with first-time teaching assistant orientation through situational comedy.

2023 Jan. **Minnesota Project in Mathematics**, Counselor  
Helped with workshops and mentored two undergraduate projects during the week-long program.

2019 – 2020 **AMS Graduate Student Blog**, Staff Writer  
Wrote about finding community through mathematical art and history.

2018 Jul. **Girls Who Code @ Cloudflare**, Workshop Leader  
Instilled an appreciation for mathematics in high school girls using elliptic cryptography puzzles.

2016 – 2017 **Mathematics Undergraduate Student Association**, UC Berkeley, President  
Helped build an inclusive and diverse community among undergraduate math students at Berkeley.

2014 & 2015 **Berkeley mini Math Tournament**, Grader and Lecturer  
Instilled an appreciation for knot theory in advanced elementary and middle school students.

## WORKSHOPS and SUMMER SCHOOLS (selected)

- 2023 Sep. **Syzygies and mirror symmetry**, American Institute of Mathematics, Pasadena, California  
Topics: resolutions of the diagonal for toric varieties, homological mirror symmetry
- Jul. **Géométrie Algébrique en Liberté**, University of Warwick, UK  
Topics: Mori Dream Spaces and quiver GIT, klt singularities, the geometry of curves
- Jun. **MRC Derived Categories, Arithmetic and Geometry**, AMS  
Topics: Frobenius pushforwards and  $F$ -thickness of the blowup  $\mathrm{Bl}_5\mathbf{P}^2$
- May **MSRI/SLMath Summer School in Commutative Algebra**, University of Notre Dame  
TA for mini-course on the geometry of nonstandard syzygies by Daniel Erman
- 2022 Dec. **RTG Workshop on Birational Complexity**, SCGP, Stony Brook University  
Topics: rationality, curves in algebraic varieties, the Cremona group, measures of irrationality
- Jun. **Pan-American School in Commutative Algebra**, CIMAT, Mexico  
Topics: positive characteristic methods, toric varieties, DG algebras, modules of differentials
- 2021 Spr. **Combinatorial Algebraic Geometry**, ICERM  
Topics: Schubert varieties, toric varieties, tropical varieties, cluster algebras and varieties
- 2018 May **RTG Summer School in Commutative Algebra**, University of Utah  
Topics: limits in positive characteristic, symbolic powers, differential operators, and syzygies.
- Mar. **Geometry of Redistricting Workshop**, University of San Francisco  
Topics: gerrymandering, voting rights, discrete geometry and graph theory

## OPEN SOURCE DEVELOPMENT

Since 2017 **Macaulay2 Internals**

I have contributed to various internal components of Macaulay2, including the engine, interpreter, core mathematical routines, and documentation. I have also contributed to the following packages:

Since 2023 **DirectSummands**: for decomposing modules and coherent sheaves, with D. Mallory.

Since 2023 **Varieties**: for computations involving projective varieties, including complexes of coherent sheaves and their morphisms, with D. Mallory, R. Ramkumar, G. Smith, K. VandeBogert.

Since 2020 **NormalToricVarieties**: added support for pullbacks of coherent sheaves over toric maps, and computations on the Cox ring of toric varieties whose class group has torsion.

Since 2021 **Truncations**: added support for truncations of modules with respect to arbitrary cones, for instance on simplicial toric varieties where the nef and effective cones differ.

Since 2020 **Saturation**: improved and added to core routines for computing annihilators, saturations, and quotients of ideals and modules, with J. Chen and M. Stillman.

Since 2019 **FGLM**: for computing Gröbner bases of zero-dimensional ideals, with D. Peifer.

Since 2018 **VirtualResolutions**: see paper above, with A. Almousa, J. Bruce, and M. Loper.

Since 2017 **LocalRings**: for symbolic computations over local rings, with M. Stillman.

## OTHER WORK & RESEARCH EXPERIENCE

2018 Summer **Cloudflare, Inc.** Cryptography Engineering Intern

Launched products involving Tor, Keyless SSL, and distributed randomness generation.

2017 Winter **Proton Research, Inc.** Cryptography Research and Development Intern

Launched support for elliptic curve cryptography in OpenPGP.js.

2017 Summer **Institute for Quantum Computing**, University of Waterloo

Topics: quantum error correcting codes from algebraic curves under John Watrous.

2015 – 2016 **Berkeley Quantum Information and Computation Center**, UC Berkeley

Topics: control theory & entanglement generation under Leigh Martin and Birgitta Whaley.

2014 Summer **Institute for Quantum Information and Matter**, Caltech

Topics: quantum game theory and semi-definite optimization under Thomas Vidick.

2013 Summer **Jet Propulsion Laboratory**, NASA

Topics: secure multiparty computation and secret sharing systems under Ed Chow.

## REFERENCES

Christine Berkesch  
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Advisor

David Eisenbud  
de@berkeley.edu

Michael Stillman  
mes15@cornell.edu

Craig Westerland  
cweste1@umn.edu  
Teaching